

Now consider

$$\phi: Y \rightarrow \mathbb{P}^1(\mathbb{F}) \setminus \mathbb{P}^1(\mathbb{F}_q)$$

$$(x, y) \mapsto [x: y]$$

this makes sense since $Y(\mathbb{F}_q) = \emptyset$

This satisfies

• For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$,

$$\phi(g(x, y)) = \phi(ax+by, cx+dy) = [ax+by: cx+dy] = g[x: y]$$

$$\phi(F(x, y)) = \phi(x^q, y^q) = [x^q: y^q] = F[x: y]$$

$$\text{• For } \alpha \in \mu_{q+1}, \phi(\alpha(x, y)) = \phi(\alpha x, \alpha y) = [\alpha x: \alpha y] = [x: y]$$

Prop $\phi: Y \rightarrow \mathbb{P}^1(\mathbb{F}) \setminus \mathbb{P}^1(\mathbb{F}_q)$ is the quotient of Y by μ_{q+1} .

Proof We need to check

1. ϕ is surj
2. $\phi(P) = \phi(Q) \Leftrightarrow Q = \alpha P$ for some $\alpha \in \mu_{q+1}$
3. $D_P \phi$ is surj for some $P \in Y$.

For any $[x: y] \in \mathbb{P}^1(\mathbb{F}) \setminus \mathbb{P}^1(\mathbb{F}_q)$, we have

$$xy^q - yx^q \neq 0 \text{ since } \left(\frac{x}{y}\right)^q \neq \frac{x}{y}$$

Then we can scale x, y to assume $xy^q - yx^q = 1 \Rightarrow \phi$ is surjective.

And $\phi(x, y) = \phi(x', y') \Leftrightarrow \exists \alpha \in \mathbb{F}^\times$ such that

$$x' = \alpha x \text{ and } y' = \alpha y$$

$$\begin{aligned} \text{But } 1 &= x'y'^q - y'x'^q \\ &= \alpha^{q+1}(xy^q - yx^q) \\ &= \alpha^{q+1} \Rightarrow \alpha \in \mu_{q+1} \end{aligned}$$

To compute $D\phi$, note that

$$\phi: Y \rightarrow \mathbb{P}^1(\mathbb{F}) \setminus \mathbb{P}^1(\mathbb{F}_q) \subseteq \mathbb{A}^1(\mathbb{F}) \text{ is } (x, y) \mapsto [x: y] \mapsto \frac{x}{y}$$

So $D\phi = \left(\frac{1}{y} \quad -\frac{x}{y^2}\right)$ and this is $\neq (0, 0)$ and has trivial kernel at any $(x, y) \in Y$. (check) $\square$

Recall $Y/G \cong \mathbb{A}^1$

$$Y/U \cong \mathbb{A}^1 \setminus \{0\}$$

$$Y/\mu_{q+1} \cong \mathbb{P}^1 \setminus \mathbb{P}^1(\mathbb{F}_q)$$

Since $Y(\mathbb{F}_q) = \emptyset$, $Y^F = \emptyset$. More generally if $\alpha \in \mu_{q+1}$,

$$Y^{\alpha F} = \emptyset$$

Indeed any $P \in Y^{\alpha F}$ would define a point on

$$(Y/\mu_{q+1})^F = (\mathbb{P}^1(\mathbb{F}) \setminus \mathbb{P}^1(\mathbb{F}_q))^F = \emptyset$$

Prop For any $\alpha \in \mu_{q+1}$

$$|Y^{\alpha F^2}| = \begin{cases} 0 & \text{if } \alpha \neq -1 \\ q^3 - q & \text{if } \alpha = -1 \end{cases}$$

Proof Let $(x, y) \in Y^{\alpha F^2}$, so

$$x = \alpha x^{q^2} \text{ and } y = \alpha y^{q^2} \text{ and } xy^q - yx^q = 1$$

$$\begin{aligned} \text{Then } 1 &= (xy^q - yx^q)^q = x^q y^{q^2} - y^q x^{q^2} \\ &= -\alpha^{-1}(xy^q - yx^q) \\ &= -\alpha^{-1} \end{aligned}$$

So $Y^{\alpha F^2} = \emptyset$ unless $\alpha = -1$. If this is the case, we want to

$$\text{solve } \begin{aligned} x &= -x^{q^2} & (1) \\ y &= -y^{q^2} & (2) \end{aligned}$$

$$xy^q - yx^q = 1 \quad (3)$$

So (x, y) satisfy (1) and (3), so

$$xy^q - yx^q = 1 \Rightarrow y^q = \frac{1+yx^q}{x}$$

$$\Rightarrow y^{q^2} = \frac{1+y^q x^{q^2}}{x^q}$$

$$y^{q^2} = \frac{1 - xy^q}{x^q} = -\frac{yx^q}{x^q} = -y$$

So we need to solve (1) and (3).

There are $q^2 - 1$ solutions to $x^q = -x$ with $x \neq 0$

For any fixed such x , there are q solutions in y to

$$xy^q - yx^q = 1$$

$$\Rightarrow |Y^{-F^2}| = q(q^2 - 1) \quad \square$$

Remark $Y^{-F^2} = \gamma^{-1}(0)$ where $\gamma: Y \rightarrow \mathbb{A}^1$ is the quotient of Y by G .

Finally, $\bar{Y}$ has a compactification

$$\bar{Y} = \{[x: y: z] \in \mathbb{P}^2(\mathbb{F}) : xy^q - yx^q = z^{q+1}\}$$

We can extend the action of G, μ_{q+1}, F to $\bar{Y}$ by

$$\bullet F([x: y: z]) = [x^q: y^q: z^q]$$

$$\bullet g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G, g[x: y: z] = [ax+by: cx+dy: z]$$

$$\bullet \alpha \in \mu_{q+1}, \alpha[x: y: z] = [\alpha x: \alpha y: z]$$

Prop $\bar{Y}$ is the closure of Y in $\mathbb{P}^2(\mathbb{F})$. It is smooth and stable under G, μ_{q+1}, F .

Proof The closure claim is standard, and the equivariance is easy. Let's check smoothness. We saw prev that Y is smooth. We need to check $\bar{Y}$ is smooth

at any $P \in \bar{Y} \setminus Y$. Note

$$\begin{aligned}\bar{Y} \setminus Y &= \{[x:y:z] \in \mathbb{P}^2(\mathbb{F}) : \\ &\quad xy^q - yx^q = 0\} \\ &= \left\{ [x:y:0] \in \mathbb{P}^2(\mathbb{F}) : y \neq 0 \text{ and } \left(\frac{x}{y}\right)^q = \left(\frac{x}{y}\right) \right\} \\ &\quad \sqcup \{[1:0:0]\} \\ &= \mathbb{P}^1(\mathbb{F}_q)\end{aligned}$$

and this identification is equiv for the G -action. Since G acts transitively on $\mathbb{P}^1(\mathbb{F}_q)$, we are reduced to showing $\bar{Y}$ is smooth at $[1:0:0]$. We use the affine nbhd

$$\mathbb{A}^2(\mathbb{F}) \cong \{x \neq 0\} \subseteq \mathbb{P}^2(\mathbb{F})$$

in which $\bar{Y}$ is defined by

$$y - y^q - z^{q+1} = 0$$

The tangent space at $(y,z) = (0,0)$ is given by

$$\varepsilon y - (\varepsilon y)^q - (\varepsilon z)^{q+1} = 0 \text{ where } \varepsilon^2 = 0$$

$$\varepsilon y = 0$$

The tangent space is $\{y=0\} \subseteq \mathbb{F}^2 = \text{Tan}_{(0,0)} \mathbb{A}^2$, is 1-dim.

$\sum \bar{Y}$ is smooth at $[1:0:0]$. $\square$

Prop The map $\bar{\pi}: \bar{Y} \rightarrow \mathbb{P}^1(\mathbb{F})$

$$[x:y:z] \mapsto [x:y]$$

is a quotient of $\bar{Y}$ by μ_{q+1} and is G and F -equiv.

Proof Similar to before. $\square$